

The Fréchet Distance between Multivariate Normal Distributions

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The Fréchet distance between two multivariate normal distributions having means μ_X , μ_Y and covariance matrices Σ_X , Σ_Y is shown to be given by $d^2 = |\mu_X - \mu_Y|^2 + \text{tr}(\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2})$. The quantity d_0 given by $d_0^2 = \text{tr}(\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2})$ is a natural metric on the space of real covariance matrices of given order.

1. INTRODUCTION

In [1], M. Fréchet introduced a metric on the space of probability distributions on R having first and second moments. The Fréchet distance $d(F, G)$ between two distributions F and G is defined by

$$d^2(F, G) = \min_{X, Y} E |X - Y|^2 \quad (1)$$

where the minimization is taken over all random variables X and Y having distributions F and G , respectively. The bivariate distribution H which minimizes the right-hand side of (1) is well-known [1, 2] to be the singular distribution with distribution function

$$H(x, y) = \min[F(x), G(y)], \quad (2)$$

where F and G are the distribution functions of F and G , respectively. In the particular case when F and G belong to a family of distributions which is closed with respect to changes of location and scale, the Fréchet distance takes the simple form

$$d^2(F, G) = (\mu_X - \mu_Y)^2 + (\sigma_X - \sigma_Y)^2 \quad (3)$$

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where μ_X, μ_Y and σ_X, σ_Y are the respective means and standard deviations of F and G .

Definition [1] generalises in an obvious way to define a metric on the space of probability distributions on R^n having second moments. The solution (2) does not apply in the case when X and Y are vectors and the evaluation of the Fréchet distance is extremely difficult in general. Again, however, the distance $d(F, G)$ is easy to determine when F and G belong to a family of n -dimensional distributions which is closed with respect to linear transformations of the random vector. We shall prove that in this case, the Fréchet distance is given by

$$d^2(F, G) = |\mu_X - \mu_Y|^2 + \text{tr}[\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2}] \quad (4)$$

where μ_X, μ_Y and Σ_X, Σ_Y are the respective means and covariance matrices of F and G , and the positive square root is taken. The above formula holds in particular when F and G are normal distributions on R^n . Additionally it will be seen that

$$d_0^2(\Sigma_X, \Sigma_Y) = \text{tr}[\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2}] \quad (5)$$

defines a metric on the space of all covariance matrices of order n .

2. MAIN RESULT

Since the results (4) and (5) generalise easily to the complex case, it will be convenient to take X and Y as complex-valued random vectors in C^n having, initially, zero means. We prove the following theorem.

THEOREM. *Let X, Y be random vectors taking values in C^n and having zero means and covariance matrices Σ_X, Σ_Y respectively. Then*

$$\text{tr}[\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2}] \leq E|X - Y|^2 \leq \text{tr}[\Sigma_X + \Sigma_Y + 2(\Sigma_X \Sigma_Y)^{1/2}] \quad (6)$$

where the square roots are the positive roots.

The bounds in (6) are attained when $X - Y$ has covariance matrix

$$\Sigma_{X-Y} = \Sigma_X + \Sigma_Y \pm [(\Sigma_X \Sigma_Y)^{1/2} + (\Sigma_Y \Sigma_X)^{1/2}], \quad (7)$$

and, for non-singular Σ_X , this occurs when X and Y are related by

$$Y = \pm \Sigma_X^{-1}(\Sigma_X \Sigma_Y)^{1/2} X. \quad (8)$$

Proof. Let W be the random vector taking values in C^{2n} defined by $W = \begin{bmatrix} x \\ y \end{bmatrix}$, and denote the covariance matrix of W by

$$\Sigma_W = \begin{bmatrix} \Sigma_X & V \\ V^* & \Sigma_Y \end{bmatrix}$$

where $*$ denotes conjugate transpose. Clearly,

$$E |X - Y|^2 = \text{tr}[\Sigma_X + \Sigma_Y - V - V^*]$$

so that we require extreme values of $\text{tr}(V + V^*)$ subject to the condition that Σ_W is a covariance matrix. We note that $\text{tr}(V + V^*)$ is a linear functional defined on a convex region of the Σ_W space and hence we can apply the method of Lagrange multipliers. Let Q denote the Hermitian form $w^* \Sigma_W w$ so that

$$Q = x^* \Sigma_X x + x^* V y + y^* V^* x + y^* \Sigma_Y y.$$

Since Q is non-negative definite, it must be capable of being written

$$Q = \sum_{i=1}^m |a_i^* x + b_i^* y|^2,$$

where $a_i, b_i \in C^n$ for $i = 1, \dots, m$ and $m \leq 2n$.

It follows that Σ_X , Σ_Y and V must have simultaneous representations of the form

$$\Sigma_X = \sum_{i=1}^m a_i a_i^*, \quad \Sigma_Y = \sum_{i=1}^m b_i b_i^*, \quad V = \sum_{i=1}^m a_i b_i^*. \quad (9)$$

We seek to minimize $\text{tr} \sum_{i=1}^m (a_i b_i^* + b_i a_i^*)$ subject to the matrices $\sum_{i=1}^m a_i a_i^*$ and $\sum_{i=1}^m b_i b_i^*$ having given values Σ_X and Σ_Y , respectively. Introducing Hermitian matrices A and M of Lagrange multipliers we seek unconstrained extreme values of the real quantity

$$S = \text{tr} \sum_{i=1}^m (a_i b_i^* + b_i a_i^*) + \text{tr} \left(\sum_{i=1}^m a_i a_i^* \right) A + \text{tr} \left(\sum_{i=1}^m b_i b_i^* \right) M.$$

It is easily seen that extreme values satisfy the conditions

$$\begin{aligned} b_i &= A a_i \\ a_i &= M b_i \end{aligned} \quad (i = 1, \dots, m).$$

Thus, A is a Hermitian matrix which by virtue of conditions (9) must satisfy the equations

$$A \Sigma_X A = \Sigma_Y \quad (10)$$

and

$$V = \Sigma_X A, \quad (11)$$

which together imply $V^2 = \Sigma_X \Sigma_Y$.

For non-singular Σ_X we may put

$$A = \Sigma_X^{-1/2} R \Sigma_X^{-1/2} \quad (12)$$

so that R is Hermitian and, because of (10), R^2 simplifies to the non-negative definite $\Sigma_X^{1/2} \Sigma_Y \Sigma_X^{1/2}$. If the latter has eigenvalues λ_i and eigenvectors s_i with $s_i^* s_j = \delta_{ij}$, then all its Hermitian square roots are of the form $\sum_{i=1}^m \varepsilon_i \lambda_i^{1/2} s_i s_i^*$ where, for each i , $\varepsilon_i^2 = 1$. From (11) and (12), $V = \Sigma_X^{1/2} R \Sigma_X^{-1/2}$ has the same eigenvalues as R , so the maximum and minimum of $\text{tr } V$, viz., $\pm \sum_{i=1}^m \lambda_i^{1/2}$, are attained when

$$V = \pm \Sigma_X^{1/2} (\Sigma_X^{1/2} \Sigma_Y \Sigma_X^{1/2})^{1/2} \Sigma_X^{-1/2}.$$

The latter product is a way of expressing $(\Sigma_X \Sigma_Y)^{1/2}$, the positive square root of $\Sigma_X \Sigma_Y$, when Σ_X is non-singular. (More generally $\Sigma_X \Sigma_Y$ has the same non-negative eigenvalues λ_i as $\Sigma_X^{1/2} \Sigma_Y \Sigma_X^{1/2}$ above, and its positive square root is $\sum_{i=1}^m \lambda_i^{1/2} u_i t_i^*$ where t_i, u_i are left and right eigenvectors of $\Sigma_X \Sigma_Y$ with $t_i^* u_j = \delta_{ij}$.) Thus, $\text{tr}(V + V^*)$ is maximised or minimised by taking

$$V = \pm (\Sigma_X \Sigma_Y)^{1/2} \quad (13)$$

That the covariance matrix of $X - Y$ is given by (7) and that relation (8) between X and Y yields the upper and lower bounds is now routine. Moreover, by continuity considerations (13) carries over to the case where Σ_X is singular; consequently (6) and (7) hold generally and the theorem is proved.

For singular Σ_X , (8) obviously no longer applies; but if the null space of Σ_X is contained in that of Σ_Y , it holds with Σ_X^{-1} replaced by the pseudo-inverse of Σ_X .

COROLLARY. *Let $\mathcal{C}_n(\Sigma)$ be the set of all covariance (non-negative, Hermitian) matrices of order n . Then*

$$d_0(\Sigma_X, \Sigma_Y) = [\text{tr}(\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2})]^{1/2} \quad (14)$$

defines a metric on $\mathcal{C}_n(\Sigma)$.

In the particular case when the real covariance matrices Σ_X and Σ_Y have the same principal axes, the metric has the particularly simple form

$$d_0^2(\Sigma_X, \Sigma_Y) = \sum_{i=1}^n (\sigma_i - \rho_i)^2 \quad (15)$$

where σ_i, ρ_i are the standard deviations of the (principal) components of X and Y , respectively, along the i th principal axis.

3. FRÉCHET DISTANCE BETWEEN MULTINORMAL DISTRIBUTIONS

Since multinormal distributions are determined completely by their means and covariance matrices, it follows immediately that the Fréchet distance between normal distributions F and G on R^n with means μ_X, μ_Y and (real) covariance matrices Σ_X, Σ_Y is given by

$$d^2(F, G) = |\mu_X - \mu_Y|^2 + \text{tr}(\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2}). \quad (16)$$

Indeed, the above result holds for any two distributions from a family of real, elliptically contoured distributions having finite second-order moments and with density function of the form

$$p(x; \mu, A) = (\text{const}) \times f((x - \mu)^* A (x - \mu)), \quad (17)$$

where f is a non-negative function on the positive real axis such that $0 < \int_0^\infty r^{n/2-1} f(r) dr < \infty$, and A is a positive-definite, symmetric matrix. In particular, by taking

$$\begin{aligned} f(u) &= \text{const}, & 0 \leq u \leq c \\ &= 0 & \text{elsewhere,} \end{aligned} \quad (18)$$

we see that Eq. (16) also determines the Fréchet distance between uniform distributions on two ellipsoids centered at μ_X, μ_Y and having covariance matrices Σ_X, Σ_Y . (This implies that the ellipsoids have the same shape and orientation as the principal ellipsoids of Σ_X, Σ_Y .) It is worth emphasizing that for non-singular Σ_X the minimizing transformation (8) between X and Y is (a) linear and (b) Hermitian. This latter property has the interpretation that the transformation is a pure strain and does not involve a rotation. This non-rotational property holds in a certain sense even if F and G have arbitrary continuous density functions in R^n provided that the minimum of $E|X - Y|^2$ is achieved by a differentiable transformation between X and Y . By consideration of nearly uniform distributions over small ellipsoids, one may show the non-rotational property of the minimizing transformation to hold locally. This result can also be proved by a standard calculus of variations argument in which $E|X - Y|^2$ is minimized subject to the constraint that the Jacobian $\partial(y)/\partial(x)$ satisfies the equation

$$\frac{\partial(y)}{\partial(x)} = \frac{p_X(x)}{p_Y(y)} \quad (19)$$

where $p_X(x)$, $p_Y(y)$ are the given density functions of the vectors X and Y , respectively.

4. TWO TRACE INEQUALITIES FOR REAL COVARIANCE MATRICES

The result

$$\min_{X,Y} E |X - Y|^2 = \text{tr}[\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{1/2}] \quad (20)$$

when X , Y have zero means and covariance matrices Σ_X , Σ_Y has two simple consequences, First,

$$\text{tr}(\Sigma_X \Sigma_Y)^{1/2} \leq \text{tr} \left(\frac{\Sigma_X + \Sigma_Y}{2} \right) \quad (21)$$

with equality iff $\Sigma_X = \Sigma_Y$. This well-known result is a generalisation of the fact that the geometric mean of two positive numbers is less than or equal to the arithmetic mean. Second, by replacing Y in (20) by tY where t is any complex number we see that

$$\min_{X,Y} E |X - tY|^2 = (\text{tr } \Sigma_Y) |t|^2 - 2 \text{tr}(\Sigma_X \Sigma_Y)^{1/2} \cdot |t| + \text{tr } \Sigma_X \geq 0$$

which implies

$$\text{tr}(\Sigma_X \Sigma_Y)^{1/2} \leq [\text{tr } \Sigma_X \cdot \text{tr } \Sigma_Y]^{1/2} \quad (22)$$

where equality holds iff $\Sigma_X = |t|^2 \Sigma_Y$ for some complex scalar t . Inequality (22) is a generalisation of the Schwarz inequality. The quantity

$$\rho = \text{tr}(\Sigma_X \Sigma_Y)^{1/2} [\text{tr } \Sigma_X \cdot \text{tr } \Sigma_Y]^{-1/2} \quad (23)$$

is the largest correlation coefficient possible between two random vectors X , Y having prescribed covariance matrices Σ_X , Σ_Y .

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